

Design and Development of a Real-Time Intelligent Vision-and-Voice-Based Navigation System using Deep Learning and Large Language Models

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ABSTRACT

Visually impaired individuals face significant challenges navigating dynamic environments due to the limitations of traditional assistive tools such as white canes, ultrasonic sensors, and GPS systems, which lack context-awareness and real-time adaptability. Similarly, drivers in low-visibility conditions like fog or headlight failure face increased risks with limited technological support. This highlights the need for a unified, intelligent, and affordable system that supports both human and vehicular navigation through real-time environmental perception and auditory feedback. However, building such a system presents key challenges: ensuring accurate, low-latency detection on resource-constrained hardware; incorporating both spatial and temporal cues for dynamic obstacle tracking; and generating intuitive, platform-independent voice alerts. Most existing systems fail to address these aspects collectively. To address these challenges, we developed a smart voice-assisted obstacle detection and navigation system that integrates lightweight deep learning for visual recognition, temporal modelling for motion detection, and Large Language Models (LLM) based speech generation for personalized feedback delivered through a low-cost portable design. First, at its core, the system employs the state-of-the-art YOLOv10 deep learning model, capable of identifying over 80 object categories including pedestrians, vehicles, and road structures with 97% detection accuracy and real-time processing speeds of 30–60 frames per second on standard hardware. Next, to improve responsiveness and contextual understanding in dynamic scenes, a background subtraction algorithm based on the Mixture of Gaussians (MOG2) algorithm is employed. The proposed work enables robust spatio-temporal feature extraction by analyzing foreground movement across sequential video frames, allowing for enhanced detection of moving objects and scene transitions. This layer of temporal intelligence complements the spatial accuracy of YOLOv10, resulting in a more adaptable and intelligent navigation system. Finally, the detection outcomes are seamlessly translated into auditory cues using Large Language Models (LLM) a hierarchical Text-to-Speech (TTS) architecture that combines Google TTS for high-quality expressive voice synthesis, Windows SAPI for offline functionality, and eSpeak for lightweight cross-platform deployment. However, the real-time voice feedback is managed by the pygame.mixer framework, ensuring non-blocking audio playback, dynamic volume adjustment, and sequential message queuing. Hardware implementation is placed on an ESP32 microcontroller, selected for its low power consumption and wireless communication capabilities. The proposed embedded sensors collection includes 360-degree LiDAR for high-resolution obstacle detection, six-axis Inertial Measurement Units (IMU) for fall detection, motion tracking and orientation estimation, and optimized power management modules that support long-duration operation with rechargeable battery systems. This sensor fusion architecture ensures millimetre-accurate spatial awareness, resulting in context-sensitive alerts such as “Person detected ahead,” “Obstacle on left,” or “Clear path confirmed”. Beyond its primary role as a mobility aid for the visually impaired, the system holds significant potential for automotive safety applications, particularly during adverse weather, nighttime driving without headlights, or tunnel navigation. By integrating advanced object detection, spatio-temporal motion analysis, intelligent audio synthesis, and multi-sensor data fusion, this work delivers a scalable, real-time solution that addresses critical gaps in current assistive and vehicular safety technologies, contributing meaningfully to accessible innovation and human-centered design.